GSSS INSTITUTE OF ENGINEERING AND TECHNOLOGY FOR WOMEN,MYSURU

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

MLOps ASSIGNMENT – 2

TRACKING A PRE-TRAINED SENTIMENT MODEL WITH MLFLOW

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1. Which run had the lowest latency, and why might that be?

The run_batch8 had the lowest latency of 0.065 seconds, followed by run_batch4 at 0.074 seconds and run_batch1 at 0.254 seconds. A larger batch size allows the model to process multiple sentences together instead of processing them one by one. This reduces processing overhead and improves inference speed.

2. Did accuracy change between runs? Why or why not?

No, the accuracy remained the same at 1.0 (100%) in all three runs. Each run used the same pre-trained sentiment analysis model and the same 10 test sentences. Only the batch size was changed. Therefore, the predicted sentiment labels remained the same, and the accuracy did not change.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

1. Reproducibility and record-keeping: MLflow automatically stores experiment parameters, metrics, and artifacts. This helps us review previous experiment results whenever needed.

2. Easy comparison: MLflow provides a user-friendly dashboard to compare different experiment runs side by side. We can easily compare accuracy, latency, and other metrics using tables and visualizations.

1. AIM To use a pre-trained sentiment analysis model and track its inference experiments using MLflow. The experiments are performed with different batch sizes (1, 8 and 4), and their parameters, metrics and prediction results are recorded and compared.

2. SOFTWARE REQUIREMENTS • Operating System: Windows • Programming Language: Python 3.10 • IDE: Visual Studio Code • Libraries: MLflow, Transformers, PyTorch, Pandas and NumPy • Pre-trained model: distilbert-base-uncased-finetuned-sst-2-english

3. INTRODUCTION

MLOps (Machine Learning Operations) is a set of practices used to manage machine learning models and their workflows. It helps organise experiments, record results and make model development and deployment easier to manage. MLflow is an open-source platform used to track machine learning experiments. It can record parameters (settings used in an experiment), metrics (measured results) and artifacts (files produced during an experiment). Its user interface allows different runs to be viewed and compared. In this assignment, a pre-trained DistilBERT

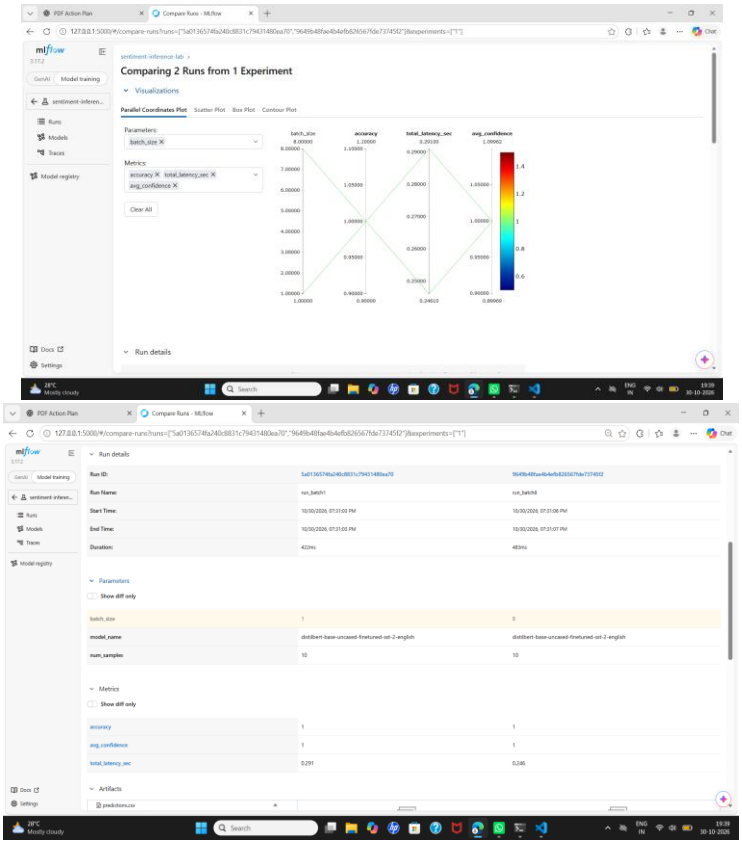
sentiment analysis model is used. The model classifies a sentence as POSITIVE or NEGATIVE. No model training is performed; only inference is carried out on ten labelled sentences. 4. PROCEDURE 1. Create a project folder and open it in Visual Studio Code. 2. Create and activate a Python 3.10 virtual environment. 3. Install MLflow, Transformers, PyTorch, Pandas and NumPy. 4. Load the pre-trained DistilBERT sentiment analysis model. 5. Prepare ten sentences with their actual sentiment labels. 6. Run inference using batch sizes 1, 8 and 4. 7. For each run, log the model name, batch size and number of samples as parameters. 8. Calculate and log accuracy, average confidence and total inference latency as metrics. 9. Save the input sentences, actual labels, predicted labels and confidence scores in predictions.csv

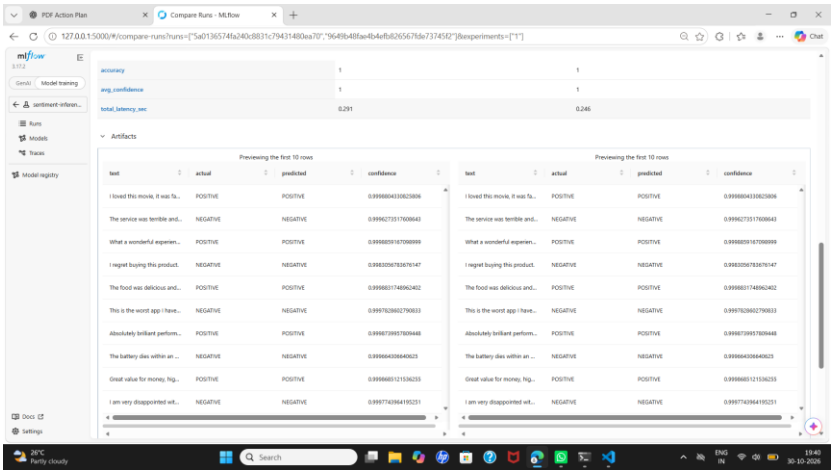
7. EXPERIMENTAL RESULTS

The following values are from the latest successful execution recorded during the experiment. The latency can vary slightly between executions because it depends on system load and runtime conditions.

Run	Batch size	Accuracy	Inference latency (sec)
run_batch1	1	1.00 (100%)	0.254
run_batch8	8	1.00 (100%)	0.065
run_batch4	4	1.00 (100%)	0.074

Note: The experiment was executed more than once, so MLflow contains duplicate runs for each batch size. The table above shows one set of latest recorded results for comparison.





10. CONCLUSION

The pre-trained DistilBERT sentiment analysis model was successfully used to perform sentiment inference with batch sizes 1, 8 and 4. MLflow recorded the experiment parameters, metrics and predictions.csv artifact. The runs were viewed and compared using the MLflow UI. In the latest execution, batch size 8 gave the lowest inference latency while maintaining 100% accuracy.